

Intergenerational Housing Misallocation and Fertility

Compact Analytical Model

Draft note

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Abstract

This note develops a compact non-spatial theory of intergenerational housing allocation and fertility. Young households decide whether to enter the modeled housing market and, conditional on entry, choose tenure, housing, consumption, and fertility. Rental and owner housing differ in their size menus, and young households face borrowing and payment constraints when they try to buy owner housing. Old households are incumbent owners who choose consumption, how much of their current owner housing to keep, and bequests. The model defines a competitive equilibrium and a constrained planner, then compares marginal allocation rules for family-capable housing.

Overview. The compact model first describes the economy, the rental and owner housing segments, and the household problems. It then defines equilibrium and the planner's problem. The main allocation result concerns constrained access to family-capable housing among entered young households. Some young households are unconstrained; others cannot upsize because rental size menus, down-payment constraints, or payment-to-income constraints bind. The old-side force, when present, is an incumbent property-tax discount: old homeowners pay less tax if they stay in their current house than if the same house were taxed at current market value. This can amplify the allocation gain from releasing owner housing, but it is not necessary for the young-access mechanism.

Remark 1 (Status). *This standalone file contains only the compact analytical model. It is copied from the v4 draft and revised to focus on tenure choice, rental-owner size segmentation, constrained access to family-capable housing, and the associated efficiency results.*

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Part I

Compact Analytical Model

1 Environment

The economy is populated by two overlapping generations of households, young and old. Young households are potential parents. They decide whether to enter the modeled housing market and, conditional on entry, choose tenure, housing, consumption, and fertility. Old households are incumbent owners. They are no longer fertile and choose how much of their current owner housing to keep, how much to consume, and how much to leave as bequests.

There is one consumption good and two housing segments. A household can rent housing services in the rental segment, denoted R , or occupy housing in the owner segment, denoted O . The two segments differ in the size menus they provide. Let

$$\mathcal{H}^R = \{h : 0 < h \leq \bar{h}^R\}, \quad \mathcal{H}^O = \{h : 0 < h \leq \bar{h}^O\},$$

with

$$0 < \bar{h}^R < \bar{h}^O.$$

Thus there is rental-owner size segmentation: large family-sized units are more available in the owner segment than in the rental segment. Housing services are divisible within each segment; $\bar{h}^R$ and $\bar{h}^O$ are individual occupancy caps, not aggregate stocks.

A renter pays flow rental cost $q^R h$. An owner pays flow user cost $q^O h$ and faces owner asset price P^O . The owner asset price is

$$P^O = \frac{q^O}{\rho_p}, \quad \rho_p = r + \delta^O + \tau^p, \quad (1)$$

where r is the interest rate, δ^O is owner-housing depreciation, and τ^p is a recurring property tax. For a fixed owner user cost q^O , a higher τ^p lowers the asset price and hence lowers the down payment required to buy a given amount of owner housing.

Let $\bar{H}^R$ and $\bar{H}^O$ denote inherited rental and owner-segment stocks. New net construction in segment $m \in \{R, O\}$ is $I^m \geq 0$, so total segment stocks are

$$H^m = \bar{H}^m + I^m, \quad m \in \{R, O\}. \quad (2)$$

Inherited owner-segment stock is consistent with old owner endowments. Let $H_{\text{pass}}^O \geq 0$ be owner-segment stock held by passive owners. Then

$$\int H_j^0 dF_O(j) + H_{\text{pass}}^O = \bar{H}^O.$$

The inherited rental stock is held by passive landlords. Passive housing income and fiscal accounts are rebated lump-sum according to the transfer rule.

Construction uses the consumption good. Costs are

$$\Phi^m(I^m; Z^m), \quad \Phi_I^m(I^m; Z^m) > 0, \quad \Phi_{II}^m(I^m; Z^m) > 0, \quad m \in \{R, O\},$$

where Z^m shifts the ease of construction in segment m . Competitive construction implies, for interior construction,

$$q^R = \Phi_I^R(I^R; Z^R), \quad q^O = \Phi_I^O(I^O; Z^O). \quad (3)$$

Housing payments are transfers across households, old owners, passive landlords, and fiscal accounts. Lump-sum rebates from these accounts are included in disposable resources. The aggregate resource cost of housing expansion is the cost of net construction, not the value of the whole existing stock.

Remark 2 (User-cost and asset-price normalization). *The compact model measures q^m as the flow user cost and marginal resource cost of one unit of segment- m housing services. Thus construction costs Φ^m are in flow units, and interior construction satisfies $q^m = \Phi_I^m(I^m; Z^m)$. The owner asset price $P^O = q^O/(r + \delta^O + \tau^p)$ only converts the owner flow cost into the purchase price relevant for collateral and down-payment constraints. If instead construction costs were measured in asset units, the owner construction condition would be $P^O = \Phi_I^O(I^O; Z^O)$, and property taxes would also distort owner construction. The flow normalization is used below.*

Let

$$x = 1 - \tau^b$$

denote the after-tax bequest share. The rule assigning estate-tax revenue or rebates is part of fiscal policy. The compact model keeps the bequest loop reduced form: old bequest expenditure is a current resource use, while young inheritance exposure B_i is a primitive of the young type distribution. Thus the estate tax affects young pledgeable resources through $A_i(x)$, but it does not introduce an additional old-side bequest incentive in this compact version.

2 Young households

There is a mass $\bar{M} > 0$ of potential young households. A potential young household has type

$$i = (y_i, a_i, B_i),$$

drawn from G_Y . Here $y_i > 0$ is disposable nonhousing income, $a_i \geq 0$ is liquid wealth, and $B_i \geq 0$ is gross inheritance exposure. Effective liquid wealth for purchase constraints is

$$A_i(x) = a_i + xB_i + T_i^b(x),$$

where $T_i^b(x)$ is the estate-tax rebate or transfer assigned to type i . The object $A_i(x)$ is pledgeable liquid wealth for purchase constraints. It is not an additional flow resource in the branch budget unless it is annuitized into y_i . Thus inheritances affect owner access through down-payment capacity rather than directly financing current consumption.

Children require consumption and housing services. Let C_i denote total nonhousing consumption expenditure. If a young household has fertility $n_i > 0$, the adult residual consumption and housing services are

$$c_i = C_i - \chi_i n_i > 0, \quad s_i = h_i - \kappa n_i > 0,$$

where $\chi_i > 0$ is the consumption requirement per child and $\kappa > 0$ is the housing requirement per child. Preferences over total nonhousing consumption have the Stone–Geary form

$$U_i^Y(C_i, h_i, n_i) = \log(C_i - \chi_i n_i) + \alpha \log(h_i - \kappa n_i) + \beta \log n_i, \quad \alpha, \beta > 0.$$

Equivalently, using adult residual consumption $c_i = C_i - \chi_i n_i$, the branch problems below use the residual utility

$$u_i^Y(c_i, h_i, n_i) = \log c_i + \alpha \log(h_i - \kappa n_i) + \beta \log n_i$$

and put the child consumption requirement in the budget. In a branch with user cost q^m , the equal-scaling benchmark is $\chi_i = \kappa q^m / \alpha$: one child then uses consumption and housing in the same proportions as the adult residual Cobb–Douglas bundle in that branch. Because $\beta \log n_i$ implies $n_i > 0$ for every entered household, the compact model studies the intensive margin of completed fertility, not childlessness.

Conditional on entry, household i chooses tenure. The renter branch is

$$W_i^R(q^R) = \max_{c_i, h_i, n_i} \{ \log c_i + \alpha \log(h_i - \kappa n_i) + \beta \log n_i \} \quad (4)$$

subject to

$$c_i + q^R h_i + \chi_i n_i = y_i, \quad (5)$$

$$h_i \in \mathcal{H}^R, \quad (6)$$

$$h_i - \kappa n_i > 0.$$

The owner branch is

$$W_i^O(q^O, P^O, x) = \max_{c_i, h_i, n_i} \{ \log c_i + \alpha \log(h_i - \kappa n_i) + \beta \log n_i \} \quad (7)$$

subject to

$$c_i + q^O h_i + \chi_i n_i = y_i, \quad (8)$$

$$(1 - \phi) P^O h_i \leq A_i(x), \quad \phi \in (0, 1), \quad (9)$$

$$q^O h_i \leq \psi y_i, \quad \psi \in (0, 1), \quad (10)$$

$$h_i \in \mathcal{H}^O,$$

$$h_i - \kappa n_i > 0.$$

Using $P^O = q^O / \rho_p$, owner housing is bounded above by

$$h_i \leq H_i^O(q^O, \tau^p, x) = \min \left\{ \bar{h}^O, \frac{A_i(x) \rho_p}{(1 - \phi) q^O}, \frac{\psi y_i}{q^O} \right\}.$$

The renter analog is $H_i^R = \bar{h}^R$. We call H_i^m the effective family-housing cap in branch m .

The inside value of entry is

$$W_i^Y(q^R, q^O, P^O, x) = \max\{W_i^R(q^R), W_i^O(q^O, P^O, x)\}. \quad (11)$$

Let $o_i \in \{R, O\}$ denote an optimal tenure.

Remark 3 (Rental premium formulation). *The cap in (6) is a compact representation of size segmentation. A smooth formulation lets renters pay*

$$R(h) = q^R h + \varrho(h),$$

where $\varrho_h(h) \geq 0$ for large units. Then the renter marginal conditions below replace q^R by $R_h(h)$.

3 Old households

Old households are indexed by j and distributed according to F_O . An old household has income $m_j > 0$, initial owner housing H_j^0 , and a warm-glow bequest utility function $\mathcal{B}_j$, possibly equal to zero.

Old household j chooses consumption c_j^O , how much current owner housing to keep, h_j^O , and bequest expenditure $e_j \geq 0$. Preferences are

$$u_j^O(c_j^O, h_j^O, e_j) = \log c_j^O + \gamma \log h_j^O + \mathcal{B}_j(e_j), \quad \gamma > 0.$$

The baseline young-access mechanism below does not require any old-side force. When the old side is active, the compact model uses a simple incumbent tax discount. Old owner j pays property taxes on a protected assessment if she stays in her current house. The same unit taxed at current market value would carry tax cost $\tau^p P^O$. The protected tax cost is $\tilde{\tau}_j \tilde{P}_j$. The difference is the per-unit tax discount from staying put:

$$L_j^\tau(q^O, \tau^p) = \tau^p P^O - \tilde{\tau}_j \tilde{P}_j \geq 0.$$

Assume

$$0 \leq L_j^\tau(q^O, \tau^p) < q^O.$$

Thus old owner j feels a private carrying cost $q^O - L_j^\tau$ per unit of housing that she keeps. Her budget is

$$c_j^O + e_j + [q^O - L_j^\tau(q^O, \tau^p)] h_j^O = m_j + q^O H_j^0, \quad 0 \leq h_j^O \leq H_j^0. \quad (12)$$

The object L_j^τ is not a preference. It is not old utility, and it is not a real resource saving. It is a tax discount attached to the incumbent owner's current house. The planner counts true old housing utility and true warm-glow bequest utility, but it does not count this tax discount as a social benefit.

4 Entry

A potential young household can enter the modeled housing market or take an outside option. The outside value is $\bar{W}^E$. Entry is subject to Type-I extreme-value taste shocks with scale $\kappa_E > 0$. The probability that type i enters is

$$\pi_i^E(q^R, q^O, P^O, x) = \frac{\exp\{W_i^Y(q^R, q^O, P^O, x)/\kappa_E\}}{\exp\{W_i^Y(q^R, q^O, P^O, x)/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}. \quad (13)$$

The endogenous measure of young entrants is

$$dF_Y^E(i; q^R, q^O, P^O, x) = \bar{M} \pi_i^E(q^R, q^O, P^O, x) dG_Y(i).$$

The limit $\kappa_E \downarrow 0$ gives deterministic entry. Conditional on entry, the household chooses tenure according to (11).

5 Competitive equilibrium

The equilibrium object is aggregate market clearing. Households take prices as given, choose tenure and housing, and their choices aggregate into segment demands. Construction and inherited stocks determine segment supplies. There is no pairwise assignment of old units to young households in the competitive equilibrium.

Given policies (τ^p, x) , construction shifters (Z^R, Z^O) , inherited stocks $(\bar{H}^R, \bar{H}^O)$, and transfer rules, aggregate rental demand at candidate prices is

$$H^{R,D} = \bar{M} \int \pi_i^E \mathbf{1}\{o_i = R\} h_i^R dG_Y(i),$$

and aggregate owner-segment demand is

$$H^{O,D} = \bar{M} \int \pi_i^E \mathbf{1}\{o_i = O\} h_i^O dG_Y(i) + \int h_j^O dF_O(j).$$

The corresponding segment supplies are

$$H^{R,S} = \bar{H}^R + I^R, \quad H^{O,S} = \bar{H}^O + I^O,$$

with construction governed by the marginal cost conditions in (3).

Definition 1 (Competitive equilibrium). *Given $(\tau^p, x, Z^R, Z^O, \bar{H}^R, \bar{H}^O)$, a competitive equilibrium consists of user costs (q^R, q^O) , owner asset price P^O , construction (I^R, I^O) , segment stocks (H^R, H^O) , young branch allocations, tenure choices, entry probabilities, old allocations, and fiscal transfers such that:*

1. *conditional on entry, young households solve (4) and (7), then choose tenure by (11);*
2. *old households solve (12);*
3. *entry is given by (13);*
4. *the owner asset price satisfies (1);*
5. *segment stocks satisfy (2) and competitive construction satisfies (3);*
6. *aggregate rental and owner-segment markets clear,*

$$H^{R,D} = H^{R,S} = \bar{H}^R + I^R, \tag{14}$$

$$H^{O,D} = H^{O,S} = \bar{H}^O + I^O; \tag{15}$$

7. *passive housing-income, property-tax, estate-tax, and policy-instrument accounts are closed by lump-sum transfers.*

Let $\Xi^\Pi \geq 0$ denote real resource costs of policy instruments. The transfer rule is chosen so that, after summing household budgets and using (14)–(15), the aggregate goods constraint is

$$\begin{aligned} & \bar{M} \int \pi_i^E [\mathbf{1}\{o_i = R\}(c_i^R + \chi_i n_i^R - y_i) + \mathbf{1}\{o_i = O\}(c_i^O + \chi_i n_i^O - y_i)] dG_Y(i) \\ & + \int (c_j^O + e_j - m_j) dF_O(j) + \Phi^R(I^R; Z^R) + \Phi^O(I^O; Z^O) + \Xi^\Pi \leq 0. \end{aligned}$$

Housing quantities are disciplined separately by segment feasibility. The goods constraint charges only net construction costs, and the old endowment term $q^O H_j^O$ is a claim on the inherited owner stock rather than new output.

Aggregate fertility is

$$N = \bar{M} \int \pi_i^E [\mathbf{1}\{o_i = R\} n_i^R + \mathbf{1}\{o_i = O\} n_i^O] dG_Y(i).$$

6 Constrained planner

The planner has the same preferences, type distributions, entry technology, inherited segment stocks, construction technologies, and rental-owner size menus as the competitive economy. The planner chooses entry-relevant inside allocations, tenure assignments, young consumption, housing and fertility, old consumption, housing and bequests, net construction, and policy instruments. The planner respects segment feasibility and the rental and owner menus.

The planner counts true old housing utility and true warm-glow bequest utility. It does not count L_j^T as utility or as a resource saving. The planner also cannot relax a down-payment or payment-to-income constraint merely with a notional transfer unless that transfer, payment subsidy, credit guarantee, or contract intervention is in the feasible instrument set and its real cost is included in Ξ^Π .

Let $\omega_i^Y > 0$ and $\omega_j^O > 0$ be Pareto weights. Let $\mathcal{G}_i(n_i)$ denote an optional social payoff from births beyond parental utility. The fertility-neutral planner has $\mathcal{G}_i \equiv 0$.

If the planner assigns young type i tenure $o_i^P \in \{R, O\}$ and allocation (c_i, h_i, n_i) , the inside value relevant for entry is

$$V_i^P = u_i^Y(c_i, h_i, n_i).$$

The induced entry probability is

$$\pi_i^P = \frac{\exp\{V_i^P / \kappa_E\}}{\exp\{V_i^P / \kappa_E\} + \exp\{\bar{W}^E / \kappa_E\}}.$$

The corresponding expected entry value, up to an additive constant, is

$$\mathcal{W}_i^E(V_i^P) = \kappa_E \log [\exp\{V_i^P / \kappa_E\} + \exp\{\bar{W}^E / \kappa_E\}].$$

The planner solves

$$\max \bar{M} \int \omega_i^Y \mathcal{W}_i^E(V_i^P) dG_Y(i) + \bar{M} \int \pi_i^P \mathcal{G}_i(n_i) dG_Y(i) + \int \omega_j^O u_j^O(c_j^O, h_j^O, e_j) dF_O(j) \quad (16)$$

subject to segment feasibility,

$$\begin{aligned} \bar{M} \int \pi_i^P \mathbf{1}\{o_i^P = R\} h_i dG_Y(i) &\leq \bar{H}^R + I^R, \\ \bar{M} \int \pi_i^P \mathbf{1}\{o_i^P = O\} h_i dG_Y(i) + \int h_j^O dF_O(j) &\leq \bar{H}^O + I^O, \end{aligned}$$

menu feasibility,

$$h_i \in \mathcal{H}^R \text{ if } o_i^P = R, \quad h_i \in \mathcal{H}^O \text{ if } o_i^P = O,$$

financial implementation, and goods feasibility. If the planner assigns type i to the owner segment and does not use an instrument that relaxes owner financial access for that type, then the assignment must also satisfy

$$(1 - \phi)P^O h_i \leq A_i(x), \quad q^O h_i \leq \psi y_i.$$

If a down-payment subsidy, credit guarantee, payment subsidy, or contract intervention relaxes these constraints, the instrument must be feasible and its real resource cost is included in Ξ^Π . Goods feasibility is

$$\begin{aligned} \bar{M} \int \pi_i^P (c_i + \chi_i n_i - y_i) dG_Y(i) \\ + \int (c_j^O + e_j - m_j) dF_O(j) + \Phi^R(I^R; Z^R) + \Phi^O(I^O; Z^O) + \Xi^\Pi \leq 0. \end{aligned}$$

Let Λ be the multiplier on goods feasibility. Let ΛQ^R and ΛQ^O be the multipliers on rental and owner segment feasibility. For a fixed tenure assignment and active set, and away from menu corners, an interior planner allocation satisfies

$$\begin{aligned} \frac{u_{h,i}^Y}{u_{c,i}^Y} &= Q^R && \text{for renter assignments,} \\ \frac{u_{h,i}^Y}{u_{c,i}^Y} &= Q^O && \text{for young owner assignments,} \\ \frac{u_{h,j}^O}{u_{c,j}^O} &= Q^O && \text{for old owner housing,} \\ Q^R &= \Phi_I^R(I^R; Z^R), \\ Q^O &= \Phi_I^O(I^O; Z^O), \\ \frac{\beta c_i}{n_i} + \nu_i^g &= \chi_i + \kappa Q_i^{o^P}, \\ \frac{\mathcal{B}'_j(e_j)}{u_{c,j}^O} &= 1 && \text{if } e_j > 0. \end{aligned}$$

The term ν_i^g is the goods-equivalent social value of a marginal birth beyond parental utility. If $\mathcal{G}_i \equiv 0$, then $\nu_i^g = 0$. Endogenous logit entry affects the levels of the first-order conditions through a common factor, but that factor cancels from the housing-consumption and fertility ratios displayed here.

7 Marginal conditions in equilibrium

The household optimality conditions imply the marginal value of relaxing the relevant housing limits and financial access constraints.

For a renter, let λ_i^R be the multiplier on (5) and $\theta_i^R \geq 0$ the multiplier on the rental size cap $h_i \leq \bar{h}^R$. At an interior allocation with respect to consumption and fertility,

$$\frac{1}{c_i^R} = \lambda_i^R,$$

$$\begin{aligned}\frac{\alpha}{h_i^R - \kappa n_i^R} &= \lambda_i^R q^R + \theta_i^R, \\ \frac{\beta}{n_i^R} &= \lambda_i^R \chi_i + \frac{\alpha \kappa}{h_i^R - \kappa n_i^R}.\end{aligned}\tag{17}$$

Define the goods-equivalent value of relaxing the rental size limit by

$$\zeta_i^R = \frac{\theta_i^R}{\lambda_i^R} \geq 0.$$

Then

$$MV_i^R \equiv \frac{u_{h,i}^Y}{u_{c,i}^Y} = \frac{\alpha c_i^R}{h_i^R - \kappa n_i^R} = q^R + \zeta_i^R.\tag{18}$$

For a young owner, let λ_i^O be the multiplier on (8), $\mu_i \geq 0$ the multiplier on the down-payment constraint (9), $\eta_i \geq 0$ the multiplier on the payment-to-income constraint (10), and $\theta_i^O \geq 0$ the multiplier on the owner menu cap $h_i \leq \bar{h}^O$. At an interior allocation with respect to consumption and fertility,

$$\begin{aligned}\frac{1}{c_i^O} &= \lambda_i^O, \\ \frac{\alpha}{h_i^O - \kappa n_i^O} &= \lambda_i^O q^O + \mu_i(1 - \phi)P^O + \eta_i q^O + \theta_i^O, \\ \frac{\beta}{n_i^O} &= \lambda_i^O \chi_i + \frac{\alpha \kappa}{h_i^O - \kappa n_i^O}.\end{aligned}\tag{19}$$

Define the goods-equivalent value of relaxing owner access by

$$\zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i^O}(1 - \phi)P^O + \frac{\eta_i}{\lambda_i^O}q^O}_{\zeta_i^{O,F}, \text{ financial component}} + \underbrace{\frac{\theta_i^O}{\lambda_i^O}}_{\text{owner-menu component}} \geq 0.$$

If the owner menu cap does not bind, then $\theta_i^O = 0$ and $\zeta_i^O = \zeta_i^{O,F}$. The young owner's marginal value of owner housing is

$$MV_i^O \equiv \frac{u_{h,i}^Y}{u_{c,i}^Y} = \frac{\alpha c_i^O}{h_i^O - \kappa n_i^O} = q^O + \zeta_i^O.\tag{20}$$

For an old owner at an interior choice of current housing to keep,

$$MV_j^O \equiv \frac{u_{h,j}^O}{u_{c,j}^O} = \frac{\gamma c_j^O}{h_j^O} = q^O - L_j^\tau(q^O, \tau^p).\tag{21}$$

If bequests are interior, then

$$\mathcal{B}'_j(e_j) = \frac{1}{c_j^O}.$$

This is a true preference margin, not an allocation distortion.

Proposition 1 (Constrained family-housing access and fertility). *Fix an entered young household and its optimal tenure branch. If the household rents, then*

$$\frac{\beta c_i^R}{n_i^R} = \chi_i + \kappa(q^R + \zeta_i^R).$$

If the household owns, then

$$\frac{\beta c_i^O}{n_i^O} = \chi_i + \kappa(q^O + \zeta_i^O).$$

Thus rental size limits and owner access constraints enter fertility through the housing services required per child.

Proof. For renters, multiply (17) by $1/\lambda_i^R = c_i^R$ and use (18). For owners, multiply (19) by $1/\lambda_i^O = c_i^O$ and use (20). $\square$

If no access or menu constraint binds in the relevant tenure branch, the corresponding access value is zero. For example, an unconstrained young owner satisfies

$$\begin{aligned} c_i^u &= \frac{y_i}{1 + \alpha + \beta}, \\ n_i^u &= \frac{\beta y_i}{(1 + \alpha + \beta)(\chi_i + \kappa q^O)}, \\ h_i^u &= \frac{\alpha y_i}{(1 + \alpha + \beta)q^O} + \kappa \frac{\beta y_i}{(1 + \alpha + \beta)(\chi_i + \kappa q^O)}. \end{aligned}$$

The renter analog replaces q^O by q^R and is feasible only if $h_i^u \leq \bar{h}^R$.

8 Efficiency

The efficiency question is whether the competitive allocation that clears the aggregate housing markets also solves the planner's allocation problem. Market clearing alone is not enough. In equilibrium, prices make aggregate rental and owner-segment demand equal aggregate supply. Efficiency requires, in addition, that the households who receive the marginal housing units are the households whose marginal values match the social marginal cost of those units.

Proposition 2 (Planner-equilibrium comparison). *Consider an interior competitive equilibrium and a planner with the same preferences, construction technologies, inherited segment stocks, and rental-owner size menus. Suppose policy accounts are resource-neutral and the planner has no social birth payoff, so $v_i^g = 0$. Suppose also that down-payment and payment-to-income constraints are private implementation constraints that the planner can relax with financing instruments whose marginal real resource cost is zero at the allocation. Conditional on the equilibrium entry measure and tenure assignments, and away from real menu corners, the competitive allocation satisfies the planner's conditional housing, fertility, and construction first-order conditions only if the financial access gap is zero for entered young owners,*

$$\zeta_i^{O,F} = 0,$$

and the old incumbent-tax-discount gap is zero,

$$L_j^\tau(q^O, \tau^p) = 0$$

for old owners at an interior choice of current housing to keep. When these conditions fail, the competitive allocation differs from the planner's allocation in the following directions:

$$MV_i^O = q^O + \zeta_i^{O,F} > q^O = Q^O$$

for a young owner constrained by down-payment or payment-to-income requirements that the planner can relax with feasible instruments, and

$$MV_j^O = q^O - L_j^\tau(q^O, \tau^p) < q^O = Q^O$$

for an old owner whose current house carries an incumbent tax discount. Conversely, if these implementation-relevant gaps are zero for all relevant households, then, conditional on entry, tenure assignments, and real menu constraints, the competitive allocation satisfies the planner's conditional housing, fertility, and construction first-order conditions for suitable Pareto weights.

Proof. Aggregate market clearing and competitive construction imply $Q^R = q^R = \Phi_I^R(I^R; Z^R)$ and $Q^O = q^O = \Phi_I^O(I^O; Z^O)$ at an interior allocation. The planner's conditional housing rules therefore require young renters away from rental-menu corners to satisfy $MV_i^R = q^R$, young owners away from owner-menu corners to satisfy $MV_i^O = q^O$, and old owners to satisfy $MV_j^O = q^O$. The competitive marginal conditions are (18), (20), and (21). Comparing these conditions gives the gaps above. The fertility conditions follow from the housing requirement per child: when the relevant access gap is zero and $\nu_i^g = 0$, the competitive fertility rules coincide with the planner's rules. Pareto weights can then be chosen to support the competitive consumption allocation. If an implementation-relevant gap is nonzero for a household with positive weight at an interior choice, the competitive allocation violates a necessary planner first-order condition. $\square$

If relaxing private financial constraints has positive marginal real resource cost, the relevant planner condition compares $\zeta_i^{O,F}$ with that cost rather than with zero. The proposition does not characterize the efficiency of the entry or tenure-assignment margins themselves; those margins are held fixed.

The broader young-access mechanism combines real family-size scarcity in the segment menus with private financial limits on owner access. The implementation-relevant gap in the proposition is the financial component $\zeta_i^{O,F}$: some young households cannot upsize into owner-segment housing services required for children because purchase constraints bind. For them, the marginal value of owner housing exceeds the segment user cost even though the owner segment clears in the aggregate. The old-side mechanism is the incumbent tax discount: an old homeowner feels carrying cost $q^O - L_j^\tau$, while the planner values the same owner-housing unit at its opportunity cost q^O . This old-side force can amplify the young-access problem, but it is not required for the young-access mechanism.

Old occupancy is not inefficient by itself. If old housing demand reflects true old housing utility or true warm-glow bequest utility, those motives are part of u_j^O and are counted by the planner. Inefficiency arises when young households face constrained access to family-capable housing, when old homeowners keep housing partly because of an incumbent tax discount, or both. The young-access result is present even when the old incumbent tax discount is zero.

Remark 4 (Real menus and instruments). *The rental-owner size menus are real constraints of the compact environment. A positive multiplier on a rental or owner menu cap is a shadow value of relaxing that real constraint, not by itself a violation of constrained efficiency when the planner faces the same menus. If the planner can use construction, conversion, assignment, or tenure-access instruments to change family-capable housing access, those instruments and their resource costs*

must appear in Ξ^Π . By contrast, down-payment and payment-to-income constraints are private financial constraints. If the planner has feasible instruments that relax them, the financial component $\zeta_i^{O,F}$ is an efficiency-relevant access gap. If the planner must instead respect the same financial constraints, those constraints must appear in the planner problem and their multipliers enter the planner's first-order conditions.

9 Fertility and entry

The fertility implication does not require a planner who directly values births. It follows from the private household problem. Children require consumption and housing services, so a constraint on family-capable housing can change fertility even when fertility enters only parental utility.

It is useful to separate a branch-level comparative static from the equilibrium comparison. Fix a tenure branch and suppress the type index. Let q be the branch user cost and let H be the effective upper bound on occupied housing in that branch. For renters, $H = \bar{h}^R$. For young owners,

$$H = \min \left\{ \bar{h}^O, \frac{A_i(x)\rho_p}{(1-\phi)q^O}, \frac{\psi y_i}{q^O} \right\}.$$

Holding $(y, \chi, q, \alpha, \beta, \kappa)$ fixed, the branch problem is

$$\max_{c,h,n} \{ \log c + \alpha \log(h - \kappa n) + \beta \log n \}$$

subject to

$$c + qh + \chi n = y, \quad h \leq H, \quad h - \kappa n > 0.$$

The unconstrained branch allocation is

$$c^u = \frac{y}{1 + \alpha + \beta}, \quad n^u = \frac{\beta y}{(1 + \alpha + \beta)(\chi + \kappa q)},$$

$$h^u = \frac{\alpha y}{(1 + \alpha + \beta)q} + \kappa \frac{\beta y}{(1 + \alpha + \beta)(\chi + \kappa q)}.$$

Suppose $0 < H < h^u$, so the housing cap binds. Then $h(H) = H$. Let

$$c(H) = y - qH - \chi n(H), \quad s(H) = H - \kappa n(H).$$

The constrained fertility choice is the unique solution to

$$\frac{\beta}{n(H)} = \frac{\chi}{c(H)} + \frac{\alpha\kappa}{s(H)}.$$

Implicit differentiation gives

$$\frac{dn}{dH} = \frac{\frac{\alpha\kappa}{s(H)^2} - \frac{\chi q}{c(H)^2}}{\frac{\beta}{n(H)^2} + \frac{\chi^2}{c(H)^2} + \frac{\alpha\kappa^2}{s(H)^2}}.$$

Therefore a relaxation of the binding housing cap raises fertility at H if and only if

$$\alpha\kappa c(H)^2 > \chi q s(H)^2.$$

This condition has a direct Stone–Geary interpretation. At an unconstrained interior branch allocation,

$$\frac{qs^u}{c^u} = \alpha.$$

The child input bundle has the same housing-to-consumption expenditure ratio as the adult residual bundle when

$$\frac{q\kappa}{\chi} = \alpha, \quad \text{equivalently} \quad \chi = \frac{\kappa q}{\alpha}.$$

More generally, fertility is strictly increasing in every binding cap relaxation if and only if

$$\chi \leq \frac{\kappa q}{\alpha},$$

so the child consumption requirement is no larger than the requirement implied by a scaled copy of the adult residual consumption-housing bundle. A binding cap implies $u_h/u_c = \alpha c(H)/s(H) > q$, so the condition above holds strictly under equal scaling whenever the cap binds. Hence, under equal scaling, every relaxation of a strictly binding family-housing cap raises fertility until the cap ceases to bind. If $\chi > \kappa q/\alpha$, fertility need not rise from every cap relaxation; in particular, $dn/dH < 0$ for binding caps sufficiently close to the unconstrained optimum. The reason is that a larger cap raises housing slack, which lowers the crowding cost of children, but it also induces the household to buy more housing at price q , reducing consumption. Fertility rises when the slack effect dominates the consumption-cost effect.

This branch result maps into the model's access margins. A relaxation of the rental size cap raises $H = \bar{h}^R$. If the owner down-payment constraint is the unique binding component of the effective owner cap, then

$$\frac{\partial H}{\partial A_i} = \frac{\rho_p}{(1 - \phi)q^O} > 0.$$

If the payment-to-income constraint is the unique binding component, then

$$\frac{\partial H}{\partial \psi} = \frac{y_i}{q^O} > 0.$$

The derivative above signs the fertility response to these primitive cap relaxations locally, holding tenure, prices, income, and child costs fixed.

The competitive-equilibrium/planner comparison is a different statement because prices, construction, consumption, tenure assignments, and entry can all change. In the competitive equilibrium, an entered young household in branch m satisfies

$$\frac{\beta c_i^{CE,m}}{n_i^{CE,m}} = \chi_i + \kappa(q^{CE,m} + \zeta_i^{CE,m}).$$

A fertility-neutral planner has no direct social birth payoff. If the planner allocation for the same type and branch is interior with respect to any remaining access constraint, then

$$\frac{\beta c_i^{P,m}}{n_i^{P,m}} = \chi_i + \kappa Q^{P,m}.$$

Hence

$$n_i^{P,m} > n_i^{CE,m} \iff \frac{c_i^{P,m}}{c_i^{CE,m}} > \frac{\chi_i + \kappa Q^{P,m}}{\chi_i + \kappa(q^{CE,m} + \zeta_i^{CE,m})}.$$

Thus a fertility-neutral planner does not mechanically deliver higher fertility merely because the competitive allocation has a positive access value. The branch-level result gives a primitive sufficient condition for a signed fertility response when a policy or planner locally raises a type's effective housing cap holding the branch primitives fixed.

If the planner directly values births through $\mathcal{G}_i(n_i)$, the interior planner condition becomes

$$\frac{\beta c_i^P}{n_i^P} + \nu_i^g = \chi_i + \kappa Q^{P,m}.$$

A positive ν_i^g is a separate fertility motive. Holding consumption, segment scarcity costs, and any remaining access constraints fixed, it raises desired fertility. In the full equilibrium, aggregate fertility also depends on prices, consumption, tenure assignments, and entry.

Let

$$n_i^Y = \mathbf{1}\{o_i = R\}n_i^R + \mathbf{1}\{o_i = O\}n_i^O$$

denote fertility in the chosen inside branch. The baseline notation normalizes fertility in the outside option to zero, so aggregate fertility among modeled entrants is

$$N = \bar{M} \int \pi_i^E n_i^Y dG_Y(i).$$

If the outside option has fertility $\bar{n}_i^E$, then aggregate fertility is instead

$$N = \bar{M} \int [\pi_i^E n_i^Y + (1 - \pi_i^E) \bar{n}_i^E] dG_Y(i),$$

and every entry-margin term below replaces n_i^Y by $n_i^Y - \bar{n}_i^E$. Across two allocations,

$$N^P - N^{CE} = \bar{M} \int [n_i^P (\pi_i^P - \pi_i^E) + \pi_i^E (n_i^P - n_i^{CE})] dG_Y(i),$$

where n_i^{CE} is fertility in the competitive-equilibrium inside branch and outside-option fertility is normalized to zero. The second term is the intensive fertility margin among the households who would enter in the competitive equilibrium. The first term is the entry margin induced by the change in inside value. If entry is held fixed, the aggregate fertility effect is just

$$N^P - N^{CE} = \bar{M} \int \pi_i^E (n_i^P - n_i^{CE}) dG_Y(i).$$

This is the aggregate analog of the planner-equilibrium comparison: the planned allocation delivers higher fertility exactly to the extent that it raises conditional fertility for constrained young types and, possibly, increases entry into the modeled housing market.

9.1 A local policy decomposition

The branch calculation gives a local fixed-price decomposition of a policy's fertility effect. Let z be a policy parameter. For renters, $H_i^R = \bar{h}^R$. For owners,

$$H_i^O = \min \left\{ \bar{h}^O, \frac{A_i(x) \rho_p}{(1 - \phi) q^O}, \frac{\psi y_i}{q^O} \right\}.$$

Let $m \in \{R, O\}$ denote the active tenure branch and let $\mathcal{B}_m(z)$ be the set of types for whom branch m is uniquely active and $H_i^m(z)$ is the unique binding family-housing cap. Define the policy pass-through to the effective cap by

$$P_i^m(z) = \frac{\partial H_i^m}{\partial z}.$$

If a policy does not relax the binding component of H_i^m , then $P_i^m(z) = 0$ for this family-space-cap mechanism. Policies that operate primarily through prices q^m require additional terms; they are not evaluated by setting $P_i^m(z) = 0$ inside this fixed-price formula.

For capped types, the fixed-branch fertility response is

$$\Psi_i^m = \frac{\frac{\alpha\kappa}{(s_i^m)^2} - \frac{\chi_i q^m}{(c_i^m)^2}}{\frac{\beta}{(n_i^m)^2} + \frac{\chi_i^2}{(c_i^m)^2} + \frac{\alpha\kappa^2}{(s_i^m)^2}}.$$

If $\chi_i \leq \kappa q^m / \alpha$ and the cap binds strictly, then $\Psi_i^m > 0$. The boundary case $\chi_i = \kappa q^m / \alpha$ is exact equal scaling.

Let

$$\Theta_i^m \equiv \frac{\partial W_i^m}{\partial H_i^m} = \lambda_i^m \zeta_i^m$$

be the utility value of relaxing the effective cap. Holding prices, tenure choices, and active sets fixed, the local aggregate fertility effect is

$$\left. \frac{dN}{dz} \right|_{q,o,A} = \bar{M} \sum_{m \in \{R, O\}} \int_{\mathcal{B}_m(z)} \left[\pi_i^E \Psi_i^m + (n_i^m - \bar{n}_i^E) \frac{\pi_i^E (1 - \pi_i^E)}{\kappa_E} \Theta_i^m \right] P_i^m(z) dG_Y(i),$$

where $\bar{n}_i^E = 0$ under the baseline outside-option normalization.

Corollary 1 (Equal scaling and access gaps). *Suppose $\chi_i = \kappa q^m / \alpha$ and the effective cap binds strictly. Then*

$$\Psi_i^m = \frac{\kappa}{\alpha} \frac{\zeta_i^m}{c_i^m} \Omega_i^m, \quad \Omega_i^m = \frac{\frac{\alpha}{s_i^m} + \frac{q^m}{c_i^m}}{\frac{\beta}{(n_i^m)^2} + \frac{\chi_i^2}{(c_i^m)^2} + \frac{\alpha\kappa^2}{(s_i^m)^2}} > 0.$$

Thus the same goods-equivalent access gap that raises the value of entry also governs the intensive fertility response in the fixed branch.

Proof. Under exact equal scaling, the numerator of Ψ_i^m is

$$\frac{\alpha\kappa}{(s_i^m)^2} - \frac{\chi_i q^m}{(c_i^m)^2} = \frac{\kappa}{\alpha} \left(\frac{\alpha}{s_i^m} - \frac{q^m}{c_i^m} \right) \left(\frac{\alpha}{s_i^m} + \frac{q^m}{c_i^m} \right).$$

At a binding cap, $\alpha c_i^m / s_i^m = q^m + \zeta_i^m$, so the first parenthesis equals ζ_i^m / c_i^m . Dividing by the positive denominator gives the expression above. $\square$

Thus the fixed-branch effect is exposure to a binding family-space constraint, times pass-through to the effective cap, times the fertility response, plus the induced entry response. The decomposition also holds tenure assignments fixed. With deterministic tenure choice, a policy that changes $W_i^O - W_i^R$ moves the indifference boundary $W_i^O = W_i^R$. If the type distribution has density near that boundary and fertility differs across branches, this tenure-switching margin is first order.

The formula above is therefore the fixed-tenure component of the local fertility effect. The next subsection characterizes this margin. A full general-equilibrium policy experiment also changes prices, construction, active sets, old housing choices, fiscal transfers, and possibly the resource cost Ξ^Π .

9.2 Tenure switching at the boundary

Tenure switching is not a price or general-equilibrium effect. It is already present in the fixed-price model because tenure is chosen by

$$W_i^Y = \max\{W_i^R, W_i^O\}.$$

A policy that raises the owner value relative to the renter value moves households near the boundary $W_i^O = W_i^R$ from renting to owning. This margin depends on the density of types near the boundary, so its magnitude is quantitative. Its sign depends on the price configuration.

To separate the result from the rent/own labels, fix prices and suppress the type index. Let A denote the cheaper branch and B the more expensive branch, with

$$q^A < q^B.$$

Let (c_m, s_m, n_m, h_m) denote the branch- m optimum, where $s_m = h_m - \kappa n_m$ and $m \in \{A, B\}$. Let ζ^m be the goods-equivalent value of relaxing the effective family-housing cap in branch m .

Proposition 3 (Fertility at a tenure switch). *Suppose $q^A < q^B$ and the household is indifferent across the two branches, $W^A = W^B$. Then the cheaper branch's effective cap binds with positive value, and the expensive branch provides more occupied housing:*

$$h_B > h_A.$$

If, in addition,

$$\chi \leq \frac{\kappa q^A}{\alpha},$$

then

$$c_B < c_A, \quad s_B > s_A, \quad n_B > n_A.$$

Proof. The cheaper branch cap must bind with a positive multiplier. If $\zeta^A = 0$, branch A attains the unconstrained value at price q^A . Unconstrained value is strictly decreasing in the user cost, and branch B has value at most its unconstrained value. Hence

$$W^B \leq W_{\text{unc}}^B(q^B) < W_{\text{unc}}^A(q^A) = W^A,$$

contradicting indifference. Thus $h_A = H^A$ and $\zeta^A > 0$.

Next, $h_B > h_A$. If instead $h_B \leq h_A$, the branch- B allocation would be feasible in branch A . At the lower user cost q^A , branch A could choose the same housing and fertility with strictly higher consumption, contradicting $W^A = W^B$.

It remains to sign fertility. Every branch optimum satisfies

$$\frac{\beta}{n} = \frac{\chi}{c} + \frac{\alpha\kappa}{s}.$$

Define

$$X = \frac{\chi}{c}, \quad Y = \frac{\alpha\kappa}{s}.$$

These variables are local to this subsection and unrelated to the bequest share x and income y_i . Then $n = \beta/(X + Y)$, and branch utility can be written as $K - G(X, Y)$, where K is a type constant and

$$G(X, Y) = \log X + \alpha \log Y + \beta \log(X + Y).$$

Tenure indifference is therefore $G(X_A, Y_A) = G(X_B, Y_B)$.

Along a level set of G ,

$$\text{sign} \left[\frac{d(X + Y)}{dX} \right] = \text{sign}(\alpha X - Y).$$

The ray $Y = \alpha X$ crosses each level set once. Indeed, since $G_Y > 0$, each level set is a strictly decreasing graph $Y = \varphi(X)$. Along the ray, $G(X, \alpha X) = (1 + \alpha + \beta) \log X + \text{constant}$ is strictly increasing in X . Hence fertility is single-peaked along an indifference curve, with fertility rising as the allocation moves toward the ray from the region $Y > \alpha X$. Economically, along an indifference curve, fertility is maximized where the adult residual bundle has the same proportions as the child input bundle, $s/c = \kappa/\chi$.

At a branch optimum,

$$\frac{\alpha c_m}{s_m} = q^m + \zeta^m, \quad \zeta^m \geq 0,$$

so

$$\frac{Y_m}{X_m} = \frac{\kappa(q^m + \zeta^m)}{\chi}.$$

The cheaper branch cap binds with $\zeta^A > 0$. Under $\chi \leq \kappa q^A/\alpha$ and $q^B > q^A$, both branch optima lie strictly in the region $Y > \alpha X$: branch A because $\zeta^A > 0$, and branch B because $q^B > q^A$.

Finally, moving along an indifference curve toward larger occupied housing raises X . To see this, note that $h = \kappa G_Y(X, Y)$ at any branch optimum. Let $T = X + Y$. Along a level set of G ,

$$\left. \frac{dG_Y}{dX} \right|_G = \frac{G_{XY}G_Y - G_{YY}G_X}{G_Y},$$

and

$$G_{XY}G_Y - G_{YY}G_X = \frac{\alpha}{XY^2} + \frac{\beta}{XT^2} + \frac{\alpha\beta X}{Y^2T^2} > 0.$$

Thus G_Y is strictly increasing in X along a level set. Since $h_B > h_A$, we have $X_B > X_A$. On the region $Y > \alpha X$, $X + Y$ falls as X rises toward the ray. Therefore $X_B + Y_B < X_A + Y_A$, and $n_B = \beta/(X_B + Y_B) > \beta/(X_A + Y_A) = n_A$. Because the level set is strictly decreasing, $Y_B < Y_A$. Hence $c_B = \chi/X_B < c_A$ and $s_B = \alpha\kappa/Y_B > s_A$. $\square$

The condition $\chi \leq \kappa q^A/\alpha$ is the same economic condition used above, evaluated at the lower user cost. It says that the child input bundle is not more consumption-intensive than the adult residual bundle in the cheap branch. Under this condition, a household indifferent between a cheap branch and an expensive branch has higher fertility in the expensive branch because the expensive branch trades adult residual consumption for family space.

This proposition immediately gives the three price configurations for the rent-own boundary. If $q^O > q^R$, ownership is the expensive family-space branch, and an indifferent switcher has $n_O > n_R$ under the condition above. If $q^O < q^R$, renting is the expensive branch, and the inequality reverses:

$n_R > n_O$ under the analogous condition at the lower owner user cost. If $q^O = q^R$ and the branches differ only through their effective caps, an indifferent household has the same allocation in both branches: either both caps are slack, or the effective caps coincide at the binding allocation. Hence $n_O = n_R$, and there is no discrete fertility jump at the tenure boundary.

This fixed-price accounting result is not a welfare result or a general-equilibrium comparative static. Let types live in a smooth finite-dimensional type space with continuous density $g(i)$, and suppose there are no mass points at the tenure boundary. Let

$$\Delta_i(z) = W_i^O(z) - W_i^R(z), \quad \mathcal{I}(z) = \{i : \Delta_i(z) = 0\},$$

and assume $\nabla_i \Delta_i(z) \neq 0$ on $\mathcal{I}(z)$, with unique branch optima and kinks in active constraints having zero measure. The switching component of the fixed-price fertility effect is

$$\left. \frac{dN^{\text{switch}}}{dz} \right|_{q, \mathcal{A}} = \bar{M} \int_{\mathcal{I}(z)} \pi_i^E (n_i^O - n_i^R) \frac{\partial_z \Delta_i(z)}{\|\nabla_i \Delta_i(z)\|} g(i) dS(i),$$

where $dS(i)$ is surface measure on the indifference boundary. For cap-shifting policies,

$$\partial_z \Delta_i(z) = \Theta_i^O P_i^O(z) - \Theta_i^R P_i^R(z), \quad \Theta_i^m = \lambda_i^m \zeta_i^m.$$

Corollary 2 (Owner-access policies). *Consider an owner-access policy with $P_i^O(z) \geq 0$ and $P_i^R(z) = 0$. Suppose the condition in Proposition 3 holds for almost every type on $\mathcal{I}(z)$, evaluated at the lower user cost. If $q^O > q^R$, the switching component is weakly nonnegative, so the fixed-tenure owner-access formula is a lower bound for this fixed-price fertility channel. If $q^O = q^R$, the switching component is zero. If $q^O < q^R$, the switching component is weakly nonpositive.*

Corollary 3 (Rental-menu policies). *Consider a rental-menu policy with $P_i^R(z) \geq 0$ and $P_i^O(z) = 0$. Suppose the condition in Proposition 3 holds for almost every type on $\mathcal{I}(z)$, evaluated at the lower user cost. If $q^O > q^R$, the switching component is weakly nonpositive: relaxing the rental cap raises fertility for constrained renters who remain renters, but it can also pull marginal owners back into renting. If $q^O = q^R$, the switching component is zero. If $q^O < q^R$, the switching component is weakly nonnegative.*

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